

MUC01 — Retail Category Intelligence

Team Section Ownership & Individual Reflections

1. Team Section Ownership

Team Member	Section / Responsibility
Chetan Kalagarla	Data Preparation & Quality Analysis
Medapati Leela Raja Vardhan Reddy	Exploratory Data Analysis & Trend Analysis
V. Shalini	Visualization & Business Insight Generation
Sai Surchit Das	Project Documentation & Repository Management

2. Individual Reflections

Chetan Kalagarla

Data Preparation & Quality Analysis

My part of the project was centred on preparing the retail transaction data and checking it for quality before any analysis began. I went through the structure of the dataset, looked closely at the columns and their data types, and worked out the number of transactions, products, categories, stores and cities we were dealing with. Missing values were checked carefully, and I also helped get the date-related fields ready so they could later be used for the monthly and day-of-week analysis. This part of the project taught me that solid exploratory analysis only works if the input data is reliable and properly understood in the first place. Watching how the teammate handling EDA and trend analysis turned these cleaned fields into meaningful business comparisons helped me see the real connection between data preparation and the insights that come after it. On the whole, this section gave me a much better feel for data quality, preprocessing, and why a dataset needs to be validated before any chart or business conclusion is built on top of it.

Medapati Leela Raja Vardhan Reddy

Exploratory Data Analysis & Trend Analysis

My contribution centred on exploring the dataset and pulling out the trends that mattered most for the business question at hand. I looked at how category-level revenue moved from January through June and compared which categories were growing against those that were slipping. The numbers showed strong growth in Apparel alongside a fairly sharp decline in Electronics, and I worked on interpreting these shifts so they could be explained as business insights rather than left as raw figures. From the teammate handling data preparation, I picked up how important it is to check the structure and quality of a dataset before jumping into analysis, and the visualization work showed me how much clearer results become once they are placed inside a well-chosen chart. This section sharpened my ability to work with grouped data, percentage change and trend comparisons to answer a real business question, and it made me realise that EDA is not just about spotting patterns – it is about explaining what those patterns could mean for someone in a Category Manager role.

V. Shalini

Visualization & Business Insight Generation

My focus was on turning the analysis into visuals that were easy to read and on shaping those visuals into insights that actually mean something for the business. I built and refined charts covering category performance, product-level performance, discount behaviour and the timing of sales, and used them to bring out findings such as the strongest and weakest categories, how individual products were performing, and how revenue differed across discount brackets and days of the week. Understanding the EDA and trend-analysis work before building each chart made a real difference – it meant every visual was grounded in numbers I actually understood, rather than just being decoration. The documentation side of the project also reminded me that a chart is only useful once it is tied to a clear business message. Working through this part of the project deepened my grasp of Matplotlib and Seaborn and taught me to choose chart types that make comparisons genuinely easy to follow. More than anything, it showed me that good visualization isn't about making something look attractive – it's about helping the audience understand the evidence, and what it means, as quickly as possible.

Sai Surchit Das

Project Documentation & Repository Management

My responsibility covered documenting the project and keeping the repository organised throughout. I worked on structuring the project materials, maintaining the README, and making sure the notebook, dataset and insight summary were all presented in a way that made sense inside the GitHub repository. I also took care of the final write-up covering the business findings, the limitations of the analysis, and the points that needed to be considered before any profitability decision was made using this work. Going through the analysis sections helped me understand how raw transaction data gradually turns into structured findings about categories, products, discounts and sales timing, and the visualization work showed me how much easier the final explanation becomes once it is supported by the right charts. This project gave me a clearer sense of how to organise a data-analysis project so that someone unfamiliar with it can follow the objective, the workflow, the dataset and the results without difficulty. It also reinforced that documentation matters because it explains not just what was done, but why it was done and what should be kept in mind before the results are used for a real business decision.

Prepared for MUC01 — Retail Category Intelligence. Section ownership and reflections above correspond to the actual contribution of each team member.